

EV Charging Network Analytics

Final Business Insights Report

1. Executive Summary

The analysis evaluates charging demand, revenue, customers, station performance, peak-hour behavior, payment methods, customer segments, and expansion opportunities across a synthetic EV charging network covering 2023–2025. The network shows a stable but largely flat operating pattern: annual charging volume remains around 166K sessions, while revenue is broadly stable with a slight decline. Performance varies substantially at the station level, making location-level analysis more informative than network-wide averages.

Management takeaway: the immediate analytical priority is targeted station and customer optimization rather than assuming that broad network expansion is required.

2. Key KPIs

Metric	Result
Total charging sessions	500,000
Successful sessions	464,973
Successful session rate	92.9%
Total revenue	■259.22M (~■25.92 Cr)
Total energy consumed	19.11M kWh (~19.11 GWh)
Active customers	20,000
Avg revenue / successful session	~■557
Avg energy / successful session	~38 kWh
Avg session duration	~74 minutes
Revenue / kWh	~■14.59
Sessions / active customer	25

3. Demand & Revenue Trends

Annual sessions were 166,895 in 2023, 166,805 in 2024 and 166,300 in 2025. This represents only a small decline across the three-year period. Annual revenue was approximately ■8.67 Cr in 2023, ■8.64 Cr in 2024 and ■8.61 Cr in 2025.

A recurring seasonal pattern appears around February and March. February repeatedly shows a month-over-month decline, followed by a strong March recovery. In 2025, February fell 8.92% from January and March then increased 10.26% from February.

Business implication: the network behaves more like a mature/stable demand system than a rapidly expanding one. Capacity planning should account for recurring seasonality.

4. City Performance

City	Sessions	Revenue
Hyderabad	61,278	■3.34 Cr
Kolkata	64,175	■3.23 Cr
Mumbai	59,420	■3.23 Cr

Pune	48,045	■2.33 Cr
Surat	44,237	■2.20 Cr

Kolkata records the highest session volume among the listed cities, while Hyderabad records the highest revenue. This demonstrates that session volume and monetization are not identical performance dimensions.

5. Peak-Hour Demand

Demand is concentrated in morning and evening periods. The highest observed hourly volumes include 19:00 (46,237 sessions), 08:00 (46,152), 20:00 (46,094), 10:00 (45,975), 09:00 (45,916), and 18:00 (45,799).

Business implication: charger availability, maintenance scheduling and operational support should be particularly attentive to these peak windows.

6. Customer Behavior & Segmentation

Segment	Customers	Sessions	Revenue
Individual	15,936	398,289	■20.64 Cr
Corporate	2,390	59,849	■3.11 Cr
Fleet	1,674	41,862	■2.17 Cr

Individual customers account for roughly 80% of sessions. The behavioral segmentation framework identifies 15,071 Regular customers, 4,889 Occasional customers and only 40 High Value customers. Because these categories were defined using project-specific thresholds, they should be presented as an analytical framework rather than an industry standard.

Business implication: the Occasional-to-Regular customer journey is a meaningful area for retention and engagement analysis.

7. Station Performance

Station-level performance varies significantly. Surat EV Hub 180 is the highest-scoring station in the performance-score analysis with a score of 97.18, followed by Hyderabad EV Hub 57 (88.53), Mumbai EV Hub 18 (88.22), Bengaluru EV Hub 42 (87.93), and Indore EV Hub 158 (87.28).

The performance score combines sessions (40%), revenue (40%), and success rate (20%). It is a project-specific scoring framework and should be interpreted as a prioritization tool rather than a universal KPI.

8. Expansion Analysis

The strict expansion rule required a station to be simultaneously in the top 25% for demand and bottom 25% for charger capacity. This returned zero stations.

Interpretation: the dataset does not identify an obvious station that is both extremely high-demand and extremely capacity-constrained under this particular rule. This supports a targeted approach to capacity decisions rather than a blanket expansion assumption.

9. Payment Behavior

UPI, Card and Wallet usage is highly balanced. Successful payment counts were 156,752 for UPI, 156,260 for Card and 156,454 for Wallet. No single method dominates revenue by a large margin in the analyzed data.

10. Recommendations

- **Prioritize station-level optimization:** use station performance and success-rate metrics to identify operational improvement opportunities.

- **Plan around peak windows:** align maintenance and operational support with morning and evening demand peaks.
- **Use city-level productivity:** compare volume and revenue together rather than using sessions alone to assess a market.
- **Improve customer engagement:** investigate ways to move Occasional customers toward more consistent usage.
- **Use targeted expansion criteria:** combine demand, capacity, utilization, failure rate and revenue before approving new capacity.
- **Monitor flat growth:** the stable three-year trend suggests that future growth initiatives should be supported by measurable customer or station-level drivers.

11. Limitations

- The dataset is synthetic and generated for portfolio/business-analysis practice.
- The uploaded analysis data did not include energy_consumption.csv, so energy-related checks in this report use charging_sessions.energy_kwh where applicable.
- The expansion and behavioral segmentation thresholds are project-defined analytical rules.
- The report does not claim that the observed synthetic patterns represent the real Indian EV charging market.

12. Final Conclusion

The network demonstrates strong operational scale and a stable customer base, but limited year-over-year growth. The strongest analytical opportunities are therefore at the station and customer levels: understand why station productivity varies, manage recurring peak demand, and improve engagement among less-active customers. The expansion analysis should remain evidence-driven rather than assuming that every high-demand location requires immediate capacity growth.

Prepared as a portfolio-focused SQL business analytics case study.